VEK280 Network Setup & PoseDetect Benchmark Guide

Serial console setup • Host connectivity • Preflight • Baseline benchmark • Webcam output

1. Connect to the board through serial

After powering the board, connect it through a serial connection using PuTTY/serial. On the VEK280 board, configure the network interface:

```
ip link set end0 up
ip addr flush dev end0
ip addr add 192.168.1.85/24 dev end0
```

Check:

```
ip addr show end0
ip route
```

You should see:

```
inet 192.168.1.85/24
```

2. Configure the host PC

On the host PC, configure the eno2 interface:

```
sudo ip link set eno2 up
sudo ip addr flush dev eno2
sudo ip addr add 192.168.1.52/24 dev eno2
```

Then:

```
ip addr show eno2
```

You should see:

```
inet 192.168.1.52/24
```

3. Check the route

Run:

```
ip route
```

You should have:

```
192.168.1.0/24 dev eno2 proto kernel scope link src 192.168.1.52
```

If it isn't there:

```
sudo ip route add 192.168.1.0/24 dev eno2 src 192.168.1.52
```

4. Check ping and SSH connection

After setup, check the ping to the IP. Then:

```
ssh root@192.168.1.10
```

5. Run preflight only

After entering the board, run preflight only. This should inspect the environment, model executable, image list, and available tools without running the full benchmark.

```
./zcu104_posedetect_benchmark.sh check
```

6. Run the baseline benchmark

Run the baseline after fixing the issues above. This runs three uninstrumented 60-second tests and generates a report. Review run validity and the raw logs before proceeding to profiling.

```
RUNS=3 THREADS=1 DURATION=60 \
./zcu104_posedetect_benchmark.sh baseline
```

7. Webcam command — live output on the host PC

If you want to see the live output on your host PC, first connect to the board using SSH with X11 forwarding.

On your host PC, run:

```
ssh -X root@192.168.1.10
```

Then, on the board:

```
cd /home/root/Vitis-AI/examples/vai_library/samples/posedetect
./test_video_posedetect sp_net 0
```